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(47) Arden's Theorem

5.4

If P and Q are two Regular Expressions over Σ , and if P does not contain ϵ , then the following equation in R given by $R = Q + RP$ has a unique solution i.e. $R = QP^*$

$$R = Q + RP \longrightarrow (1)$$

$$R = QP^*$$

$$= Q + QP^*P$$

$$= Q(\epsilon + P^*P)$$

$$= Q.P^*$$

Proved

(2) — unique solution

$$R = Q + RP$$

$$= Q + [Q + RP]P$$

$$= Q + QP + RP^2$$

$$= Q + QP + [Q + RP]P^2$$

$$= Q + QP + QP^2 + RP^3 \quad \dots = Q + QP + QP^2 +$$

$$+ QP^n + RP^{n+1}$$

$$[R = QP^k]$$

$$= Q + QP + QP^2 + \dots + QP^n + QP^k \cdot P^{n+1}$$

$$= Q [E + P + P^2 + \dots + P^n + P^{n+1} \cdot P^k]$$

$$= Q \cdot P^k$$

$$\boxed{R = QP^k}$$

(48) An Example Proof using Identities of Regular Expressions.

Prove that,

$(1+00^*1) + (1+00^*1)(0+10^*1)^*(0+10^*1)$ is equal to $0^*1(0+10^*1)^*$

$$LHS = (1+00^*1) [E + (0+10^*1)^*(0+10^*1)]$$

$$= (1+00^*1) \cdot (0+10^*1)^*$$

$$= (E + 00^*1) \cdot (0+10^*1)^*$$

$$= (E + 00^*)1 \cdot (0+10^*1)^*$$

$$= 0^*1 \cdot (0+10^*1)^* \quad \checkmark$$



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(49) Designing Regular Expression - Examples

over $\{a, b\}$

1) Language accepts strings of length exactly 2

$$L_1 = \{aa, bb, ab, ba\}$$

$$\Rightarrow R = aa + ab + ba + bb$$

$$= a(a+b) + b(a+b)$$

$$= \underline{(a+b)(a+b)} \checkmark$$

2) Language accepting strings of length at least 2

$$R = \underline{(a+b)(a+b)(a+b)^*}$$

$$L_2 = \{aa, ab, ba, bb, aaa, \dots\}$$

3) Language accepting of length at most 2

$$L_3 = \{\epsilon, a, b, aa, ab, ba, bb\}$$

$$R = (a+b)(a+b) + (a+b) + \epsilon$$

$$= (a+b) + (a+b + \epsilon) + \epsilon$$

$$= \underline{(\epsilon + a + b)(\epsilon + a + b)}$$